Epidemiology and Outcomes of Lisfranc Injuries Identified at the National Football League Scouting Combine

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Background: Lisfranc injuries are challenging to treat and may have a detrimental effect on athletic performance.

Purpose: (1) Determine the epidemiological characteristics of Lisfranc injuries at the annual National Football League (NFL) Scouting Combine, (2) define player positions at risk for these injuries, and (3) evaluate the impact that these injuries and radiographic findings have on NFL draft position and performance.

Study Design: Cohort study; Level of evidence, 3.

Methods: All players who sustained a Lisfranc injury prior to Combine evaluation between 2009 and 2015 were evaluated. The epidemiological characteristics, player positions affected, treatment methods, and number of missed collegiate games were recorded. Radiographic outcomes were analyzed via Combine radiograph findings, while NFL performance outcomes were assessed for all Lisfranc injuries (2009-2013) compared with matched controls in the first 2 years of play.

Results: A total of 41 of 2162 (1.8%) Combine participants were identified with Lisfranc injuries, of whom 26 of 41 (63.4%) were managed operatively. Players who underwent surgery were more likely to go undrafted compared with players managed nonoperatively (38.5% vs 13.3%, operative vs nonoperative management, respectively; $P = .04$) and featured a worse NFL draft pick position (155.6 vs 109; $P = .03$). Lisfranc-injured players when compared with controls were noted to have worse outcomes in terms of NFL draft position (142 vs 111.3, Lisfranc-injured players vs controls, respectively; $P = .04$), NFL career length 2 years or longer (62.5% vs 69.6%; $P = .23$), and number of games played (16.9 vs 23.3; $P = .001$) and started (6.8 vs 10.5; $P = .08$) within the first 2 years of their NFL career. Radiographs demonstrated that 17 of 41 (41.5%) athletes had residual Lisfranc joint displacement greater than 2 mm compared with the contralateral foot. Lisfranc-injured athletes with greater than 2 mm residual displacement, when compared with matched controls, had worse draft position (156.9 vs 111.2 for Lisfranc-injured players vs controls, respectively; $P = .009$) and fewer games played (14.4 vs 23.3; $P = .001$) and started (3.1 vs 10.5; $P = .03$). Moreover, athletes with greater than 2 mm residual displacement featured worse outcomes across all assessed NFL variables versus athletes with residual displacement of 2 mm or less.

Conclusion: Lisfranc injuries identified at the NFL Combine have an adverse effect on an NFL athlete's draft status, draft position, and overall play during initial NFL seasons. In particular, residual displacement of the Lisfranc joint has a detrimental effect on the first 2 seasons of NFL play and may lead to long-lasting negative effects on the athlete's career.

Keywords: Lisfranc; tarsometatarsal joint; football; NFL

Foot injuries account for greater than 15% of all athletic injuries,[5] with midfoot sprains alone affecting 4% of collegiate American football players each year.[10] Collegiate and professional football places high demands on the tarsometatarsal joints; therefore, injuries to this area will adversely affect performance. Lisfranc injuries may involve damage to both the ligament and bone complex and result in painful instability, which affects push-off strength and balance. Fortunately, Lisfranc injuries account for only a small percentage of all foot injuries, with an incidence of approximately 1 in 55,000 persons every year in the general population.[4] However, Lisfranc injuries are being increasingly diagnosed in athletes, particularly those participating in American football.[10]

At the annual National Football League (NFL) Scouting Combine, approximately 3% of more than 10,000 collegiate football players are invited to a 5-day program that allows all 32 NFL teams to evaluate the best collegiate talent prior to the NFL draft.[2,7] In addition to obtaining performance data on common strength and conditioning tests, each team's medical staff performs a comprehensive history and

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physical examination to determine a player's current health and to evaluate any prior injuries. To our knowledge, few data are available to assist with estimating the impact of a history of Lisfranc injury on NFL draft status and, ultimately, future NFL performance.

Lisfranc injuries may have a detrimental effect on athletic performance and, in particular, professional athletic careers. Understanding the incidence of these injuries at the NFL Combine and the subsequent impact on performance in the NFL will allow physicians to better counsel injured athletes on expectations and help determine optimal treatment. The purposes of this study are to (1) determine the incidence of Lisfranc injuries at the annual NFL Scouting Combine, (2) identify player positions at higher risk for these injuries, and (3) evaluate the impact that these injuries and their radiographic findings have on NFL draft position and performance.

METHODS

Epidemiological Characteristics

A retrospective review of all collegiate injuries identified at the NFL Combine was performed with data collected from a comprehensive medical evaluation performed on all players including history, physical examination, physician interviews, prior medical records, and imaging studies completed at the Combine. Approval was obtained from the institutional review board (Protocol #2015P002224/MGH), the NFL Players Association (NFLPA), and the NFL Physicians Society (NFLPS) Research Committee. The review included all players who had sustained a Lisfranc injury prior to participating in the NFL Combine from 2009 to 2015 identified at any point during the medical evaluation. The epidemiological characteristics of Lisfranc injuries were evaluated by recording the total number of injuries, positions affected, and treatment methods. Odds ratios were calculated to identify the risk of incurring a Lisfranc injury in specific positions. Treatment methods were analyzed by recording whether nonoperative or operative interventions were used. For those who underwent operative intervention, the type of operative fixation was described, including screw fixation, suspensory fixation, or arthrodesis. Moreover, the number and types of additional operative procedures following primary surgery were recorded. Last, the number of missed collegiate games to allow for recovery from Lisfranc injury was noted for each player.

NFL Career Impact

The NFL career impact of Lisfranc injuries on Combine participants from 2009 through 2013 was determined through analysis of NFL draft position, NFL career length 2 years or longer, and number of games played and started within the first 2 years of NFL career for each player (Appendix Table A1, available in the online version of this article). NFL performance data were collected through the use of STATS.com, which is a web-based, publicly accessible comprehensive sport statistics database. Moreover, injury history during the first 2 years of NFL career for each Lisfranc-injured player was collected through the use of Pro-Football-Reference.com (Sports Reference LLC). Players who participated in the 2014 and 2015 Combines were excluded from outcomes analysis because these data were unavailable at time of statistical analysis. Comparative analysis was performed for groups within the study cohort to evaluate variables including player position, treatment methods, and radiographic outcomes.

Players from the study cohort were then compared with position-matched controls. The control group was selected based on fulfillment of the following criteria: (1) no history of a midfoot injury, (2) no significant missed time ($\leq$2 total missed games in college), (3) no history of any prior surgery, (4) played same position. The standard practice for collection of the number of missed games during collegiate play for each NFL Combine participant is direct transfer of data from the National Collegiate Athletic Association (NCAA) injury surveillance system including number of games missed as well as the reason attributed to each missed game. This is standard and is carried out uniformly for all injuries and players. In addition, controls were similarly evaluated through analysis of draft position, career length 2 years or longer, and number of games played and started within the first 2 NFL seasons.

Radiographic Analysis

For analysis of plain radiographs, Lisfranc joint spacing was measured between the base of the first and second metatarsals on weightbearing anteroposterior (AP) foot radiographs. Grading of the residual diastasis was performed according to a modified soft-tissue Lisfranc sprain classification from Nunley and Vertullo,[13] with nondisplaced Lisfranc joints classified as grade 1, Lisfranc joint diastasis of 1 to 5 mm classified as grade 2, and greater than 5 mm diastasis

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TABLE 1
Risk for Lisfranc Injury by Position

	Lisfranc Injury, n (%)	Odds Ratio	*P*
All athletes (n = 2162)	41 (1.9)	—	—
Total offense (n = 1148)	27 (2.4)	1.7	.109
Quarterback (n = 117)	8 (6.8)	4.4	.004
Running back (n = 238)	1 (0.4)	0.2	.115
Wide receiver (n = 302)	6 (2.0)	1.1	.903
Tight end (n = 129)	4 (3.1)	1.7	.318
Offensive lineman (n = 362)	8 (2.2)	1.2	.639
Total defense (n = 1014)	14 (1.4)	0.6	.109
Defensive lineman (n = 376)	3 (0.8)	0.4	.098
Linebacker (n = 233)	6 (2.6)	1.4	.434
Defensive back (n = 405)	5 (1.2)	0.6	.306

classified as grade 3. In addition, residual Lisfranc joint displacement was calculated as the difference between the diastasis of the previously injured foot compared with the contralateral foot. Residual Lisfranc joint displacement was then stratified into 2 groups for comparison: (1) 2 mm or less residual displacement and (2) greater than 2 mm displacement. This cutoff value (2 mm) was chosen based on previous findings that have suggested a greater likelihood of a previous Lisfranc injury if a displacement greater than 2 mm is seen between the first and second metatarsal bases in comparison to the contralateral, uninjured foot.[11] Last, evidence of degenerative changes about the midfoot was noted.

Statistical Analysis

Odds ratios were calculated by use of logistic regression to assess the risk of sustaining a Lisfranc injury for each position. Two-sample, 2-tailed *t* tests were completed to assess differences in outcomes between positions (offensive vs defensive players), treatment groups (nonoperative vs operative), Lisfranc joint radiographic displacement categories (≤2 mm vs >2 mm), and cases and controls.

RESULTS

Epidemiological Characteristics

Forty-one (1.9%) of the 2162 players participating in the annual NFL Scouting Combine from 2009 to 2015 who had a previous Lisfranc injury were identified. Quarterbacks were noted to have an increased risk of sustaining a Lisfranc injury compared with other positions (odds ratio [OR] 4.4; 95% confidence interval [CI], 1.6-12.1; $P = .004$). No other significant differences in injury risk were found among individual positions or when comparing offensive to defensive positions (Table 1).

Of the players who sustained a Lisfranc injury, 15 of 41 (36.6%) players were treated nonoperatively, while 26 of 41 (63.4%) underwent operative intervention. Of those who underwent surgery, open reduction internal fixation (ORIF) with screws and/or plates was performed in 20 of 26 (76.9%) players, suspensory fixation in 5 of 26 (19.2%) players, and arthrodesis in 1 of 26 (3.8%) players (Figure 1). A subsequent procedure to remove retained hardware was performed in 14 of 26 (53.8%) athletes. No additional revision operative procedures were performed in the operative group.

Overall, the 41 athletes with a history of Lisfranc injury missed a mean ± SD 2.9 ± 4.9 collegiate football games to allow for recovery from this injury. The number of missed collegiate games was significantly higher in the operative group compared with the nonoperative group (4.3 ± 5.7 vs 0.5 ± 1.1, respectively; $P = .008$). When offensive and

Figure 1. Operative treatment of Lisfranc injuries. Weightbearing AP foot radiographs from Combine demonstrating retained hardware from prior (A) open reduction internal screw fixation (n = 20/26, 76.9%), (B) suspensory button fixation (n = 5/26, 19.2%), and (C) arthrodesis (n = 1/26, 3.9%).

TABLE 2
Nonoperative vs Operative Treatment of Lisfranc Injury

	Collegiate Games Missed Due to Lisfranc Injury, Mean (SD)	Midfoot Degenerative Joint Disease, n (%)	Percentage Undrafted	Overall Draft Position, Mean (SD)
All Lisfranc injuries (n = 41)	2.9 (4.9)	13 (31.7)	29.3	134.7 (71.0)
Nonoperative (n = 15)[a]	0.5 (1.1)	3 (20)	13.3	109 (80.7)
Operative (n = 26)[b]	4.3 (5.7)	10 (38.5)	38.5	155.6 (56.2)
t test[c]	*P* = .008	*P* = .23	*P* = .04	*P* = .03

[a]Missed games for athletes with nonoperative management: range 0-3, median 0, interquartile range 0.
[b]Missed games for athletes with operative management: range 0-24, median 2.5, interquartile range 10.
[c]Two-sample, 2-tailed *t* tests were performed to assess differences based on treatment.

TABLE 3
Impact of Lisfranc Injury on NFL Careers[a]

	n (%)	Undrafted, n (%)	Overall Pick, Mean (SD)	NFL Career Length ≥2 y, n (%)	GP, Mean (SD)	GS, Mean (SD)
Offense						
Lisfranc injury	15 (62.5)	5 (33.3)	129.8 (83.2)	9 (60)	16.4 (10)	7.4 (10.3)
Controls	346	118 (34.1)	114.4 (68.4)	220 (63.6)	22.3 (7.9)	10.5 (10.2)
P value		.47	.25	.39	.02	.18
Defense						
Lisfranc injury	9 (37.5)	3 (33.3)	162.3 (51.4)	6 (66.7)	17.5 (8.4)	5.8 (6.6)
Controls	355	81 (22.8)	108.6 (70.1)	268 (75.5)	24.2 (7.6)	10.5 (10.5)
P value		.23	.03	.28	.02	.14
All players						
Lisfranc injury	24	8 (33.3)	142 (72.8)	15 (62.5)	16.9 (9.1)	6.8 (8.8)
Controls	701	199 (29.4)	111.3 (69.3)	488 (69.6)	23.3 (7.8)	10.5 (10.4)
P value		.30	.04	.23	.001	.08

[a]GP, total number of games played in first and second years of NFL career; GS, total number of games started in first and second years of NFL career. The data exclude athletes who attended the NFL Combine in 2014 and 2015.

defensive players were compared, no significant difference in lost time following Lisfranc injury was detected (2.6 ± 5.3 vs 3.6 ± 4.2; *P* = .55).

NFL Career Impact

Overall, the 41 athletes with a history of Lisfranc injury achieved a mean NFL draft position of 134.7 ± 71.0, with 12 (29.3%) athletes going undrafted. The percentage of players undrafted (38.5% vs 13.3%; *P* = .04) and the overall draft position (155.6 vs 109; *P* = .03) were both significantly greater in the operative versus nonoperative groups, respectively (Table 2). When comparing offensive and defensive players from 2009 to 2013 (Table 3), no significant differences were detected in mean overall draft position (129.8 ± 83.2 vs. 162.3 ± 51.4; *P* = .29) and percentage undrafted (33.3% vs. 33.3%; *P* = .99)

When Combine participants from 2009 through 2013 with a history of Lisfranc injury (n = 24) were compared with matched controls, Lisfranc-injured players were noted to have worse overall performance across all assessed NFL variables, including NFL draft position (142 vs 111.3, Lisfranc-injured players vs controls, respectively; *P* = .04), NFL career length 2 years or longer (62.5% vs 69.6%; *P* = .23), and number of games played (16.9 vs 23.3; *P* = .001) and started (6.8 vs 10.5; *P* = .08) within the first 2 years of their NFL career (Figure 2). Of the 24 players, 5 (20.8%) experienced an injury during either of the first 2 years of their NFL career resulting in a missed game, with an average of 4.7 (SD = 4.5) games missed. However, none of the 24 Lisfranc-injured players missed a game during the first 2 years of their NFL career as a result of a foot reinjury.

Radiographic Analysis

Analysis of Combine plain radiographs at a mean 21 months (range, 1-54 months) from Lisfranc injury revealed 17 of 41 (41.5%) athletes with a residual Lisfranc joint displacement of more than 2 mm compared with the contralateral uninjured foot (Figure 3). Lisfranc-injured athletes with more than 2 mm residual Lisfranc joint displacement, when compared with athletes who had 2 mm or less residual displacement, featured a worse overall NFL draft position (156.9 vs 116.6, >2 mm vs ≤2 mm, respectively; *P* = .06) and lower number of NFL games started (3.1 vs 10; *P* = .06) (Table 4).

Moreover, significant differences were seen between controls and players with greater than 2 mm displacement for draft position (111.2 vs 156.9, respectively; *P* = .009)

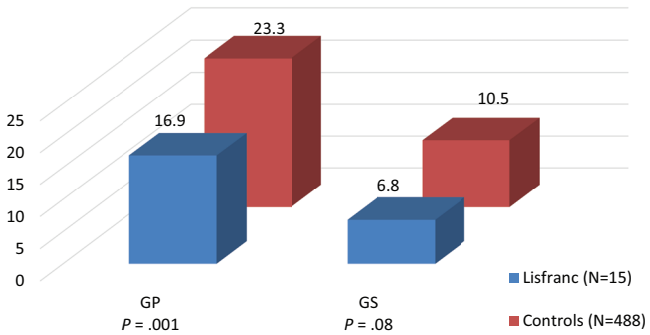

Figure 2. NFL games played and started in 2 NFL seasons: Lisfranc-injured players vs controls. When Combine participants from 2009 through 2013 with a history of Lisfranc injury were compared with matched controls, Lisfranc-injured players were noted to have fewer games played and started within the first 2 years of their NFL careers. GP, games played; GS, games started.

and games played (23.3 vs 14.4; P = .001) and started (10.5 vs 3.1; P = .03). Matched controls did not significantly differ from players with 2 mm or less displacement with respect to draft position (111.2 vs 116.6, respectively; P = .38) and games played (23.3 vs 19.0; P = .06) and started (10.5 vs 10.0; P = .44) during first 2 NFL seasons.

No significant difference was detected when comparing rates of midfoot degenerative changes between the nonoperative and operative subgroups (20% vs 38.5%, respectively; P = .23). When residual radiographic diastasis was graded without comparison to the uninjured foot (Figure 4), no significant differences among NFL performance variables were found.

DISCUSSION

Epidemiological Characteristics

Midfoot injuries have been reported to occur in up to 4% of collegiate American football players each year.[10] In our study, 41 (1.8%) of 2162 players participating in the annual

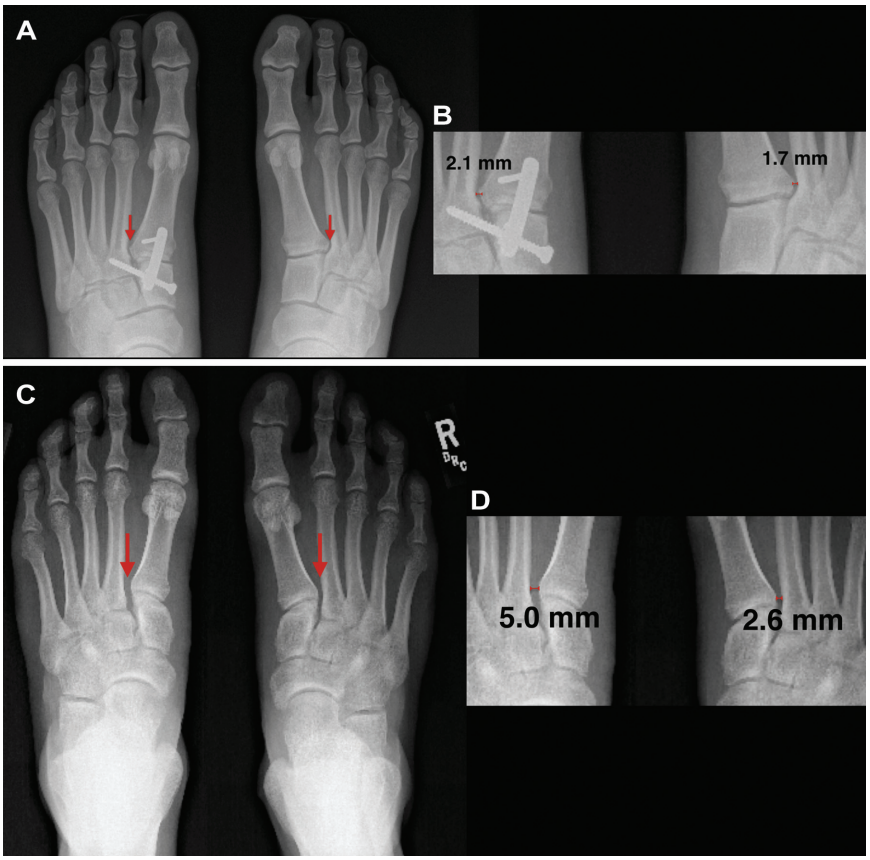

Figure 3. Classification of residual Lisfranc joint diastasis with comparison to uninjured contralateral foot. (A) Weightbearing AP foot radiographs after surgical fixation of the Lisfranc joint demonstrating less than 2 mm residual diastasis versus the contralateral foot. (B) After surgical fixation, there was 2.1 mm distance across the tarsometatarsal joint versus 1.7 mm distance in the normal contralateral foot, for a 0.4-mm side-to-side difference. (C) Weightbearing AP foot radiographs of the Lisfranc joint demonstrating greater than 2 mm residual diastasis versus the contralateral foot. (D) In the affected foot, there was 5.0 mm distance across the tarsometatarsal joint versus 2.6 mm distance in the normal contralateral foot, for a 2.4-mm side-to-side difference.

TABLE 4
Impact of Residual Lisfranc Joint Displacement on NFL Performance[a]

	n (%)	Missed Games, Mean (SD)	Undrafted, n (%)	Overall Pick, Mean (SD)	NFL Career Length ≥2 y, n (%)	GP, Mean (SD)	GS, Mean (SD)
All Lisfranc	41	2.9 (4.9)	12 (29.3)	134.7 (71)	15 (62.5)	16.9 (9.1)	6.8 (8.8)
Displacement ≤2 mm	24 (58.5)	4.2 (5.7)	8 (33.3)	116.6 (74.2)	8 (53.3)[b]	19 (8.8)[b]	10 (10.4)
Displacement >2 mm	17 (41.5)	1.2 (2.9)	4 (23.5)	156.9 (62.4)[c]	7 (77.8)	14.4 (9.5)[d]	3.1 (5.1)[e]
t test[f]		$P = .02$	$P = .25$	$P = .06$	$P = .12$	$P = .18$	$P = .06$
Controls	972	0.02 (0.2)	277 (28.5)	111.2 (69.1)	488 (69.6)	23.3 (7.8)	10.5 (10.4)

[a]GP, total games played in first and second year of NFL; GS, total number of games started in first and second year of NFL. The data exclude players drafted in 2014 and 2015 (7 cases and 701 controls).
[b-e]P values for cases relative to controls: [b]$P \leq .1$; [c]$P \leq .01$; [d]$P \leq .001$; [e]$P \leq .05$.
[f]A 2-sample, 2-tailed *t* test among cases stratified by displacement group.

Figure 4. Classification of residual Lisfranc joint diastasis without comparison to uninjured contralateral foot. Weight-bearing AP foot radiographs from Combine showing (A) a grade 1 Lisfranc joint with less than 1 mm residual diastasis and (B) a grade 3 Lisfranc joint with greater than 5 mm residual diastasis.

NFL Scouting Combine from 2009 to 2015 with a previous Lisfranc injury were identified. The prevalence of Lisfranc injuries in our cohort is notably lower than that reported by Meyer et al.[10] However, it is possible that a prior Lisfranc injury may decrease a player's likelihood of receiving an invitation to the NFL Combine.

Reports indicate that specific position players in American football may be at increased risk of sustaining Lisfranc injuries. In the study cohort of Meyer et al,[10] a particularly high percentage of offensive linemen (14.6%), defensive backs (13.5%), and linebackers/defensive ends (13.0%) sustained midfoot sprains. Although Meyer et al reported that no quarterback sustained a midfoot sprain in their cohort, our study demonstrates that players at this position may be at increased risk for this injury. However, it is possible that a history of Lisfranc injury may not decrease a quarterback's likelihood of being invited to the Combine, compared with other positions, due to the decreased emphasis placed on mobility at the quarterback position.

The reported absence from play following a Lisfranc injury has varied in the available literature from 3.1 days to 11.1 months.[3,9,10,13-15] In our study, athletes missed a mean 2.9 ± 4.9 collegiate football games due to Lisfranc injury. Time lost from athletic competition was greater for athletes in the operative group compared with the nonoperative group as measured by the number of missed collegiate games (4.3 vs 0.5, respectively; $P = .008$). However, it is likely that operative intervention implies a more severe initial injury pattern. Osbahr et al[14] reported that nonoperative management of grade 1 and 2 midfoot sprains resulted in mean 3- and 36-day absences, respectively, while operative intervention for grade 3 sprains resulted in a 73-day or greater absence from play.

NFL Career Impact

When Combine participants with a history of Lisfranc injury were compared with matched controls, players with a previous Lisfranc injury were noted to have worse overall performance on all tested NFL variables. Significant differences were identified for overall draft pick ($P = .04$) and number of games played ($P = .001$) within the first 2 years in the NFL between Lisfranc-injured athletes and controls. Moreover, only 5 of 24 (20.8%) players sustained an injury during their first 2 years of NFL career resulting in a missed game. More specifically, one of these injuries included a severe knee injury resulting in 13 games missed, which resulted in a large standard deviation of 4.5 games missed per injury. Ultimately, no foot reinjury was noted in any of the players during the first 2 years of their NFL career that led to a missed game. These data indicate that Lisfranc injuries might result in a mild decline in NFL career production.

While multiple studies[3,9,10,12-14] have reported on the rate of return to athletic competition following Lisfranc injury, only one study[9] commented on the quality of

postinjury athletic performance. McHale et al[9] reported on the offensive and defensive statistical production of NFL players who sustained Lisfranc injuries compared with matched controls. The investigators' analysis revealed declines in offensive and defensive statistical production following Lisfranc injury that did not reach statistical significance. In addition, the investigators found no significant difference in career length of Lisfranc-injured athletes compared with controls, as measured by total number of games played. These findings are consistent with the small declines in NFL draft position, NFL career length 2 years or longer, and number of games played and started within the first 2 years of NFL career that were identified in our present study.

Of note, players with Lisfranc injuries who underwent surgery in our study cohort had a greater chance of remaining undrafted ($P = .04$) and being picked at a worse overall NFL draft position ($P = .03$) compared with players treated nonoperatively. While it is possible that operative intervention may compromise an athlete's ability to perform at the professional level, it is likely that surgery may be a potential surrogate for initial injury severity. Due to lack of access to initial injury radiographs, however, it was not possible to confirm this assertion through our current study. Nevertheless, these data may still provide insight into the differential impact of the spectrum of Lisfranc injury on professional football careers.

Radiographic Analysis

Lisfranc-injured athletes with greater than 2 mm residual Lisfranc joint displacement, compared with those athletes with 2 mm or less residual displacement, were shown to have missed fewer collegiate games (1.2 vs 4.2, >2 mm vs ≤2 mm, respectively; $P = .02$) but had a worse overall NFL draft position (156.9 vs 116.6; $P = .06$) and lower number of NFL games played (14.4 vs 19; $P = .18$) and started (3.1 vs 10; $P = .06$). In addition, residual Lisfranc joint displacement was correlated with inferior performance outcomes. When residual radiographic diastasis was graded without comparison to the uninjured foot in accordance with the modified Nunley classification,[13] no significant differences among NFL performance variables were detected in our study. This finding suggests that comparison to contralateral films may provide more accurate prognostic information than the modified Nunley classification alone.

Prior studies have similarly correlated residual Lisfranc joint displacement with poor functional outcomes in both athlete and nonathlete patient populations. While only 3 of 24 (12.5%) midfoot sprains in the study by Meyer et al[10] were found to have radiographic widening between the first and second metatarsals, 1 of these 3 collegiate football players reported recurrent midfoot pain more than 1 year following injury, accounting for 25% of players with residual functional problems in their study. Nunley and Vertullo[13] reported excellent functional outcomes in 14 of 15 athletes treated for Lisfranc injuries; however, 2 patients with increased Lisfranc diastasis were unable to play their respective sports until after revision ORIF. One of these patients reported residual pain following return to football, accounting for the only treatment failure in that series.

Studying a nonathlete population, Arntz et al[1] reported that the results of ORIF procedures for Lisfranc injuries were highly dependent on the accuracy of the reduction, as "overall good or excellent" functional results were obtained in 26 of 27 patients with closed Lisfranc injuries when anatomic reduction was achieved compared with only "fair" functional results in 1 patient with residual Lisfranc joint incongruity. In addition, Myerson et al[12] reported that the quality of the initial reduction was the major determinant of treatment outcome in their study of 52 patients with 55 Lisfranc injuries. Similarly, Kuo et al[6] studied 48 patients treated for Lisfranc injuries with ORIF and reported decreased functional outcomes scores and increased rates of degenerative changes in patients with nonanatomic reductions (>2 mm displacement) compared with those with anatomic reductions on final posttreatment stress AP radiographs. Our results support these studies and provide further evidence that adequate reduction is a major predictor of future performance and outcomes.

Analysis of Combine radiographs from participants with a history of Lisfranc injury revealed 13 (31.7%) athletes with evidence of degenerative changes about the midfoot. No significant difference was detected when we compared rates of posttraumatic midfoot arthritis in the operative and nonoperative cohorts (38.5% vs 20%; $P = .23$). Prior studies have reported similar rates of posttraumatic midfoot arthritis following Lisfranc injuries, ranging from 25% to 75%.[6,8]

Limitations

This study has several limitations. First, data for this study were collected through a retrospective review of charts from the NFL Combine. It is possible that some Lisfranc injuries in this patient population may not have been discovered at this event. However, each player undergoes a rigorous physical examination with thorough histories, which are performed by the medical staff of 32 NFL teams over the course of the event, thereby decreasing the likelihood of missed injuries. Second, this study did not assess the impact that initial injury severity or the status of the repair immediately following surgery may have on NFL performance outcomes due to the unavailability of initial injury as well as immediate postoperative radiographs, which would help further distinguish the greatest determining factor in the overall outcome. Similarly, it was not possible to distinguish between purely ligamentous Lisfranc injuries and those with associated fractures because initial radiographs were unavailable. Third, this study was underpowered and unable to identify differences between outcomes of specific operative techniques due to the variability in operative constructs, which included suspensory fixation, arthrodesis, and ORIF with varying numbers of screws and plates. These limitations may be of

particular significance for the study of Lisfranc injuries due to the spectrum of severity and variable treatment options for Lisfranc injuries when compared with other athletic injuries. Further studies are needed to fully assess these variables and provide additional prognostic information regarding Lisfranc injuries in collegiate athletes. Fourth, success at the professional level is highly dependent on various factors beyond the scope of this study. However, this study attempted to control for these variables as carefully as possible through use of a matched control group.

CONCLUSION

Lisfranc injuries identified at the NFL Combine have an adverse effect on an NFL athlete's draft status, draft position, and overall play during initial NFL seasons. In particular, residual displacement of the Lisfranc joint has a detrimental impact on the first 2 seasons of NFL play and may lead to long-lasting negative effects on an athlete's career.

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